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Total No. of printed pages = 03

Monsoon, 2025

B. Tech 5th Semester Examination

Environmental Studies

Course Code: MOPEC-1

Full Marks – 50

Time – 2 hours

The figure in the margin indicates full marks for the questions.

SECTION I

(10x1=10)

1. Choose the correct answer:

- a) The organisms that feed on dead organisms, and wastes of living organisms are called as
 - (i) Chemotrophs
 - (ii) Carnivores
 - (iii) Detritivores
 - (iv) Decomposers
- b) The movement of nutrients in an ecosystem is cyclic while the flow of energy is
 - (i) Bidirectional
 - (ii) Unidirectional
 - (iii) Multidirectional
 - (v) None of the above
- c) In a grassland ecosystem, if 10,000 KJ of energy is available at the producer level, approximately how much energy will be available to tertiary consumers?
 - (i) 100 KJ
 - (ii) 10 KJ
 - (iii) 1 KJ
 - (vi) 1000 KH
- d) Western ghats are very rich in endemic species of
 - (i) Birds
 - (ii) Lions
 - (iii) Amphibians
 - (iv) Turtles

- e) What is the link between an ecosystem's biotic and abiotic Components?
- (i) Energy flow
 - (ii) Matter cycling
 - (iii) Both (i) and (ii)
 - (iv) None of the above
- f) Itai-Itai disease in Japan was caused by the consumption of rice contaminated with
- (i) Mercury
 - (ii) Iron
 - (iii) Cadmium
 - (iv) Zinc
- g) Assertion- Reason:
 Assertion(A): NPP is always less than GPP
 Reason(R): A part of GPP is used by plants for respiration?
- (i) Both A and R are true
 - (ii) Both A and R are false
 - (iii) A is true, but R is false
 - (v) A is false, but R is true
- h) Primary challenge in urban rain water harvesting
- (i) Oxides of Sulphur
 - (ii) Oxides of Nitrogen
 - (iii) Smoke
 - (iv) Carbon Monoxide
- i) Cleaner production technology focusses on
- (i) Treating pollutants after release
 - (ii) Reducing pollutants at source
 - (iii) Storing pollutants safely
 - (iv) Recycling only metals
- j) The Air (Prevention and Control of Pollution) Act was enacted to control air pollution by
- (i) Industries only
 - (ii) Vehicles only
 - (iii) All sources of emissions
 - (iv) Forest fires only

SECTION II

Answer *any one* question from this section:

Marks: 8

2. (a) Define the Ecosystem with a proper diagram. What are food chains and food webs? Give examples with their significance. (2+3+3=8)
- (b) What are ecological pyramids? Why are some pyramids upright while others are inverted in different ecosystems? What are biogeochemical cycles? (2+3+3=8)

SECTION III

Answer *any one* question from this section

Marks:8

3. (a). Compare genetic diversity, species diversity, and ecosystem diversity. Analyse Indian biodiversity with special reference as a megadiversity nation. (4+4=8)
- (b) Examine the major causes of man-wildlife conflicts. List the remedial steps that can minimize the conflict. (4+4=8)

SECTION IV

Answer *any one* question from this section

Marks:8

4. (a) Identify the different particulate matter related to air pollution. (8)
- (b). Identify how as an individual effort can be given to prevent environmental pollution. (8)

SECTION V

Answer *any one* question from this section

Marks:8

5. (a). Evaluate the role of solid waste management in urban waste control. Include case studies and modern waste minimization strategies (4+4=8)
- (b). Estimate the possible measures that can be adopted to reduce greenhouse gases in the environment. (8)

SECTION VI

Answer *any one* question from this section

Marks:8

6. (a). Assess the importance of environmental ethics in promoting sustainable development. Highlight the role of environmental laws and acts in addressing environmental issues. (4+4=8)
- (b) Evaluate the effectiveness of the Environmental Protection Act (1986) in addressing modern environmental challenges such as climate change and industrial pollution. (8)